

Supply Chain Analysis Report

Project Title

Supply Chain Analysis Dashboard using Power BI

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BATCH: 08-DEC-25

TEAM ID: PTID-CDA-APR-26-11954

Introduction

Supply chain management plays a critical role in ensuring the efficient movement of products from suppliers to customers while maintaining optimal inventory levels, controlling logistics costs, and ensuring product quality. Organizations rely on data-driven insights to monitor supply chain performance and make informed operational decisions.

The objective of this project is to analyse supply chain data and identify key factors affecting revenue generation, inventory management, transportation efficiency, and quality control. The analysis focuses on understanding product performance, stock distribution across locations, shipping costs, inspection outcomes, and overall operational effectiveness.

The dataset was extracted from a SQL database and pre-processed using Microsoft Excel before being imported into Power BI for visualization and analysis. An interactive dashboard was developed to track key performance indicators (KPIs), visualize operational trends, and generate actionable business insights.

The findings from this analysis help identify opportunities for revenue growth, inventory optimization, logistics cost reduction, and process improvement. By leveraging data analytics and business intelligence techniques, the project supports strategic decision-making and contributes to improved supply chain efficiency.

Business Problem

Effective supply chain management is essential for maintaining product availability, controlling operational costs, and ensuring customer satisfaction. However, organizations often face challenges in monitoring inventory levels, managing logistics expenses, tracking product quality, and identifying high-performing product categories.

Without a centralized analytical view, decision-makers may struggle to identify revenue drivers, optimize stock distribution, reduce transportation costs, and address operational bottlenecks. This can lead to increased costs, inventory imbalances, delayed inspections, and reduced overall efficiency.

The business problem addressed in this project is to analyze supply chain operations and identify key factors influencing revenue generation, inventory management, logistics performance, and quality control. By leveraging data analytics and interactive visualizations, the organization can gain actionable insights into operational performance and make informed business decisions.

The goal of this analysis is to answer the following business questions:

- Which product categories generate the highest revenue?
- How are inventory levels distributed across locations?
- Which transportation modes incur the highest shipping costs?
- What is the current status of product quality inspections?
- Are there operational inefficiencies that require management attention?

Addressing these questions enables the organization to improve supply chain efficiency, optimize resource allocation, reduce costs, and support long-term business growth.

Data Understanding

The dataset used in this project contains supply chain operational data related to product performance, inventory management, logistics operations, and quality control. The data was extracted from a SQL database and analysed to gain insights into key business processes that influence supply chain efficiency and profitability.

The dataset includes information on product categories, revenue generated, stock levels, lead time, shipping costs, transportation modes, inspection results, customer demographics, and geographic locations. These variables provide a comprehensive view of supply chain operations and support performance monitoring across different business functions.

Key Attributes and Business Significance

Attribute	Description	Business Significance
Product Type	Category of products	Identifies high-performing product segments
Revenue Generated	Revenue earned from products	Measures business performance and profitability
Price	Product selling price	Supports pricing and revenue analysis
Stock Levels	Inventory available at each location	Helps monitor inventory distribution and stock availability
Lead Time	Time taken to fulfil orders	Indicates operational efficiency
Shipping Costs	Cost incurred for transportation	Supports logistics cost analysis
Transportation Modes	Road, Rail, Air, Sea, etc.	Evaluates transportation efficiency
Inspection Results	Pass, Fail, Pending	Measures product quality performance
Defect Rate	Percentage of defective products	Helps assess quality control effectiveness
Customer Demographics	Customer category information	Supports customer-focused analysis
Location	Operational city/location	Enables geographical performance analysis

Dataset Summary

The dataset was extracted from a MySQL database and contains supply chain operational data related to product performance, inventory management, logistics operations, and quality control.

Dataset Characteristics:

- **Source:** MySQL Database
- **Data Type:** Structured Supply Chain Data
- **Key Areas Covered:**
 - Product Performance
 - Revenue Analysis
 - Inventory Management
 - Logistics Operations
 - Quality Control
 - Geographic Distribution

The dataset provides a comprehensive view of supply chain activities and supports performance monitoring across multiple business functions.

Data Preprocessing

Before performing the analysis, the dataset was extracted from a MySQL database and cleaned using Microsoft Excel. Since the dataset size was relatively small, Excel was used to efficiently perform data validation and preprocessing activities.

The preprocessing process involved checking the dataset for missing values, identifying duplicate records, and verifying data accuracy across all columns. Numerical fields such as Revenue Generated, Price, Stock Levels, Lead Time, Shipping Costs, and Defect Rate were reviewed to ensure consistency and correctness.

Categorical variables including Product Type, Transportation Mode, Inspection Results, Customer Demographics, and Location were standardized to maintain uniform naming conventions throughout the dataset. Data formatting issues were corrected where necessary, and location information was validated to support geographical analysis in Power BI.

After completing the cleaning and validation process, the dataset was transformed into an analysis-ready format and imported into Power BI for KPI creation, visualization development, and business insight generation.

SQL Data Extraction

The supply chain dataset was stored and managed in a MySQL database. SQL was used to retrieve, validate, and inspect the data before it was exported for further preprocessing and visualization. This step ensured that the dataset was complete, accurate, and ready for analysis.

Several SQL queries were executed to verify the dataset structure, check record availability, and extract supply chain information required for the project. Basic validation activities included row count verification, table structure inspection, and sample record extraction.

Sample SQL Queries Used

```
use project_supply_chain_data;
```

```
select *
```

```
from supply_chain_data;
```

```
select count(*) as total_rows
```

```
from supply_chain_data;
```

```
describe supply_chain_data;
```

```
select *
```

```
from supply_chain_data
```

```
limit 10;
```

The screenshot displays the MySQL Workbench interface. The 'Query' tab is active, showing a SQL script with 11 lines. The script includes a 'use' statement, a 'select *' query, a 'count(*)' query, a 'describe' statement, and another 'select *' query with a 'limit 10'. The 'Result Grid' shows the output of the 'select *' query, displaying 10 rows of data. The 'Vertical Output' section shows the output of the 'describe' statement, displaying the table structure.

Product type	SKU	Price	Availability	Number of products sold	Revenue generated	Customer demographics	Stock levels	Lead times	Order quantities	Shipping times	Shipping carriers	Shipping costs	Supply name
haircare	SKU0	69.808006	55	802	8661.996792	Non-binary	58	7	96	4	Carrier B	2.956572	Supple
skincare	SKU1	14.843523	95	736	7460.900065	Female	53	30	37	2	Carrier A	9.716575	Supple
haircare	SKU2	11.319683	34	8	9577.749626	Unknown	1	10	88	2	Carrier B	8.054479	Supple
skincare	SKU3	61.163343	68	83	7766.836426	Non-binary	23	13	59	6	Carrier C	1.729569	Supple
skincare	SKU4	4.805496	26	871	2686.505152	Non-binary	5	3	56	8	Carrier A	3.890548	Supple
haircare	SKU5	1.699976	87	147	2828.348746	Non-binary	90	27	66	3	Carrier B	4.444099	Supple
skincare	SKU6	4.078333	48	65	7823.476560	Male	11	15	58	8	Carrier C	3.880763	Supple
cosmetics	SKU7	42.958384	59	426	8496.103813	Female	93	17	11	1	Carrier B	2.348339	Supple
cosmetics	SKU8	68.717597	78	150	7517.363211	Female	5	10	15	7	Carrier C	3.404734	Supple

Vertical Output: supply_chain_data 16 x

Output: Action Output

#	Time	Action	Message	Duration / Fetch
22	12:17:08	select 'Product type', sum('Revenue generated') as Total_Revenue from supply_chain_data group by...	3 row(s) returned	0.140 sec / 0.000 sec
23	12:20:17	select * from supply_chain_data	100 row(s) returned	0.250 sec / 0.000 sec

REVENUE ANALYSIS QUERY

```
select `product type`,  
       sum(`revenue generated`) as total_revenue  
from supply_chain_data  
group by `product type`  
order by total_revenue desc;
```

The query above was used to identify the highest revenue-generating product categories and support revenue performance analysis within the Power BI dashboard.

SQL Validation Output

The extracted dataset contained product information, inventory levels, revenue metrics, transportation details, inspection outcomes, and location data. SQL validation confirmed that the dataset was complete and suitable for downstream preprocessing and dashboard development.

Figure: MySQL Workbench showing SQL query execution and supply chain dataset extraction.

The screenshot displays the MySQL Workbench interface. The left sidebar shows a list of schemas, including 'Automobile Insurance', 'customer segmentation', 'datamites_sql', 'dell-company', 'Fastfood analysis', 'OnlineBookstore', 'onlinebookstore', 'project_automobile_insurance', 'project_banking', 'project_crop_production_india', 'project_csat_analytics', 'project_customer_segmentation', 'project_food', 'project_health_data', and 'project_insurance'. The main editor window shows a SQL query for revenue analysis. The 'Result Grid' at the bottom displays the output of the query, showing three rows of data for 'Product type' and 'Total_Revenue'. The 'Vertical Output' section shows the execution log, indicating that the query was executed successfully and returned 3 rows.

Product type	Total_Revenue
skincare	241628.162133
haircare	174455.390606
cosmetics	161521.266001

#	Time	Action	Message	Duration / Fetch
21	12:15:33	select * from supply_chain_data limit 10	10 row(s) returned	0.110 sec / 0.000 sec
22	12:17:08	select 'Product type', sum('Revenue generated') as Total_Revenue from supply_chain_data group by ...	3 row(s) returned	0.140 sec / 0.000 sec

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to understand the distribution, trends, and relationships within the supply chain dataset before developing the dashboard. The objective was to identify key patterns related to revenue generation, inventory levels, logistics performance, and product quality.

The exploratory analysis revealed several important business patterns within the supply chain dataset. **Skincare products** generated the highest revenue contribution (**241.63K**), making them the strongest-performing product category. Inventory analysis showed that **Kolkata** maintained the highest stock level (**1,439 units**), indicating a significant concentration of inventory at a single location.

The logistics analysis identified **Road Transportation** as the most expensive transportation mode, incurring the highest shipping cost (**161**). Quality control analysis revealed that **Pending Inspections** accounted for approximately **41%** of total inspection outcomes, highlighting a potential operational backlog that requires management attention.

Additionally, the overall **Defect Rate** remained low at **2.28%**, indicating stable product quality performance, while only **one stockout incident** was recorded, reflecting effective inventory management practices. These findings helped identify key operational trends and guided the development of KPIs, dashboard visualizations, business insights, and recommendations presented in the Power BI dashboard.

Dashboard Development

An interactive Supply Chain Analysis Dashboard was developed using Power BI to provide a comprehensive view of key operational metrics and support data-driven decision-making. The dashboard was designed to enable users to monitor supply chain performance through intuitive visualizations, KPIs, and interactive filtering options.

The dashboard includes key performance indicators (KPIs) such as Total Revenue, Average Price, Average Lead Time, Average Defect Rate, and Stockout Count to provide a high-level overview of business performance.

Several visualizations were incorporated to analyse different aspects of the supply chain:

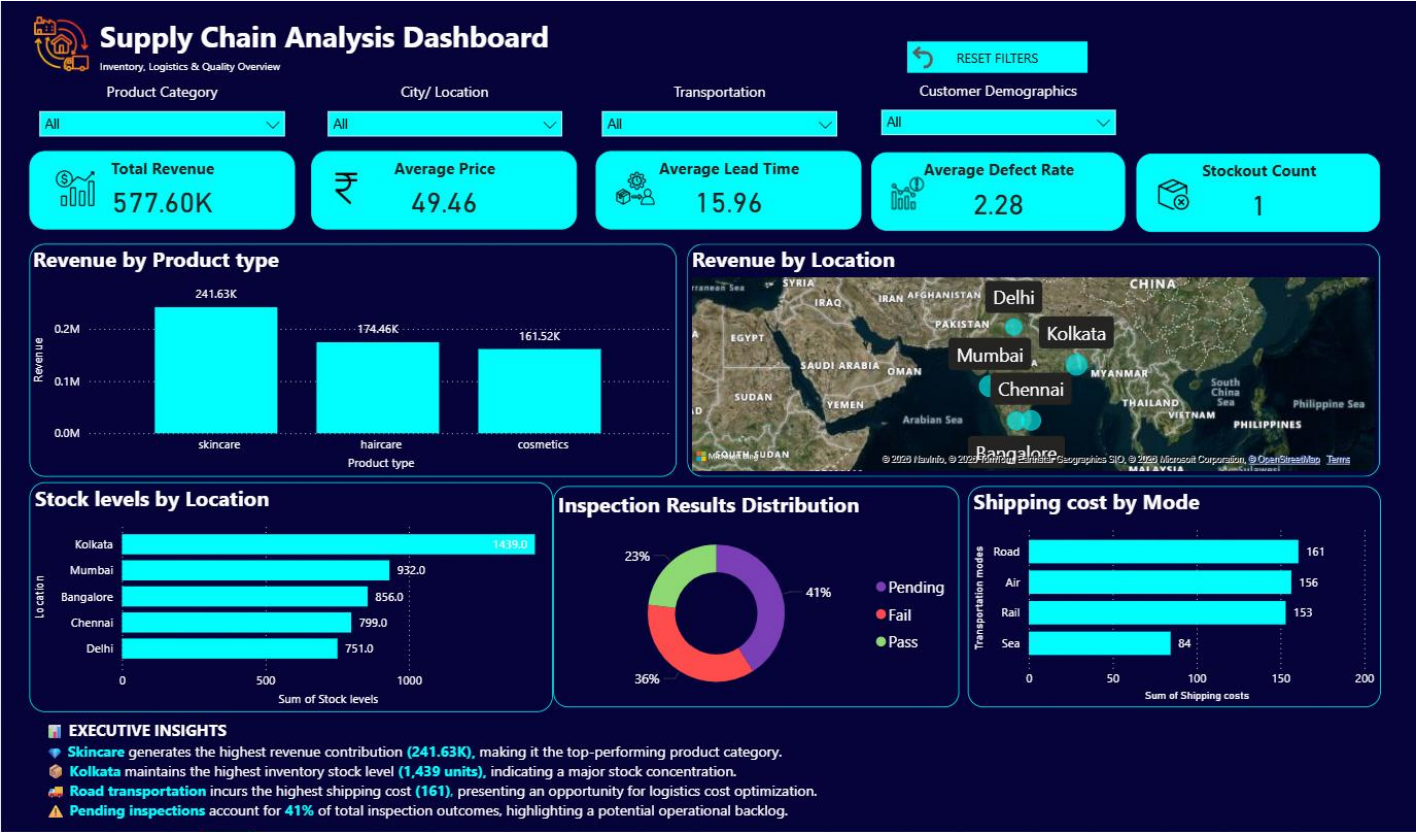
- **Revenue by Product Type** to identify top-performing product categories.
- **Revenue by Location (Map Visualization)** to analyse geographical revenue distribution.
- **Stock Levels by Location** to monitor inventory allocation across cities.
- **Inspection Results Distribution** to evaluate product quality and inspection status.
- **Shipping Cost by Transportation Mode** to assess logistics expenses and transportation efficiency.

Interactive slicers were added for Customer Demographics and Product Type, allowing users to dynamically filter and explore the data. A reset button was implemented to quickly restore the dashboard to its default state.

To enhance business storytelling, an Executive Insights section was included to highlight key findings and support management decision-making. The dashboard was designed using a consistent colour theme, clear visual hierarchy, and user-friendly layout to improve readability and usability.

The final dashboard provides a centralized view of supply chain performance and enables stakeholders to identify trends, monitor KPIs, and make informed business decisions.

Figure 1: Interactive Supply Chain Analysis Dashboard developed using Power BI.



Key Insights

Based on the analysis of the supply chain dataset and the Power BI dashboard, the following key insights were identified:

Revenue Performance

- **Skincare products** generated the highest revenue contribution (**241.63K**), making them the strongest-performing product category.
- Product category performance varied significantly, indicating opportunities to focus on **high-revenue product segments**.

Inventory Analysis

- **Kolkata** maintained the highest inventory stock level (**1,439 units**), indicating a major concentration of inventory at this location.
- Inventory distribution was uneven across locations, suggesting the need for **inventory balancing**.

Logistics Performance

- **Road transportation** incurred the highest shipping cost (**161**) among all transportation modes.
- Transportation costs varied across modes, highlighting opportunities for **logistics optimization** and **cost reduction**.

Quality Control Analysis

- **Pending inspections** accounted for approximately **41%** of total inspection outcomes, representing the largest inspection category.
- The high number of pending inspections may lead to **operational delays** and requires management attention.

Operational Performance

- The overall **Defect Rate** remained low at **2.28%**, indicating stable product quality across the supply chain.
- Only **1 Stockout Incident** was recorded, demonstrating effective inventory management and product availability.

Geographic Insights

- Supply chain operations were distributed across multiple locations, with certain cities contributing more significantly to **inventory** and **revenue performance**.
- Geographic analysis helped identify key **operational hubs** within the supply chain network.

These insights provide valuable information for improving **revenue growth**, **inventory allocation**, **logistics efficiency**, and **quality control processes**.

Business Recommendations

Based on the findings obtained from the Supply Chain Analysis Dashboard, the following recommendations are proposed to improve operational efficiency, reduce costs, and enhance overall supply chain performance:

Product Strategy

- Increase focus on **Skincare products**, as they generate the highest revenue (**241.63K**) and represent the strongest-performing product category.
- Implement targeted marketing and inventory planning strategies to maximize revenue from high-performing products.

Inventory Optimization

- Review inventory allocation across locations and reduce excessive stock concentration in **Kolkata (1,439 units)**.
- Adopt a balanced inventory distribution strategy to improve stock utilization and reduce holding costs.

Logistics Cost Reduction

- Analyse **Road Transportation**, which incurs the highest shipping cost (**161**), to identify opportunities for route optimization and cost savings.
- Evaluate alternative transportation modes and carrier performance to improve logistics efficiency.

Quality Improvement

- Prioritize the resolution of **Pending Inspections (41%)** to minimize operational delays and strengthen quality control processes.
- Implement regular monitoring and automated tracking mechanisms to improve inspection turnaround time.

Operational Efficiency

- Continue monitoring the **Defect Rate (2.28%)** to maintain product quality standards.
- Maintain current inventory control practices, as only **one stockout incident** was recorded, indicating effective stock management.

Strategic Decision-Making

- Utilize the Power BI dashboard for continuous performance monitoring and data-driven decision-making.
- Regularly review KPIs and operational metrics to identify trends, risks, and opportunities for business improvement.

Implementing these recommendations can help improve supply chain efficiency, optimize resource utilization, reduce operational costs, and support sustainable business growth.

Business Impact

The Supply Chain Analysis Dashboard provides a centralized and interactive platform for monitoring key supply chain performance indicators. By transforming raw operational data into meaningful insights, the dashboard enables stakeholders to make informed and data-driven business decisions.

The analysis identified high-performing product categories, inventory concentration across locations, logistics cost drivers, and quality control challenges. These insights help management optimize inventory allocation, reduce transportation costs, improve inspection processes, and enhance overall operational efficiency.

The dashboard also supports proactive decision-making by allowing users to monitor critical KPIs such as Revenue, Lead Time, Defect Rate, and Stockout Count in real time. Through continuous performance tracking and analysis, the organization can improve resource utilization, minimize operational risks, and support sustainable business growth.

Overall, the solution enhances supply chain visibility and provides actionable insights that contribute to improved business performance and strategic planning.

Conclusion

The Supply Chain Analysis Dashboard successfully transformed raw operational data into actionable business intelligence. By integrating **MySQL** for data extraction, **Microsoft Excel** for data preprocessing, and **Power BI** for visualization, the project provided a comprehensive analysis of revenue performance, inventory management, logistics efficiency, and quality control processes.

The dashboard enables stakeholders to identify revenue opportunities, optimize inventory allocation, monitor logistics costs, and improve operational efficiency through data-driven decision-making. The insights generated from this analysis support proactive management of supply chain operations and help organizations achieve improved performance, cost optimization, and sustainable business growth.

The analysis revealed that **Skincare products** were the highest revenue contributors, **Kolkata** maintained the largest inventory stock levels, **Road Transportation** incurred the highest shipping costs, and **Pending Inspections** represented the largest share of inspection outcomes. These findings highlight important areas for operational improvement and strategic decision-making.

The dashboard enabled effective monitoring of critical KPIs such as **Total Revenue**, **Average Price**, **Lead Time**, **Defect Rate**, and **Stockout Count**, providing stakeholders with a clear understanding of overall supply chain performance.

Based on the insights generated, recommendations were proposed to optimize inventory allocation, reduce logistics costs, improve quality control processes, and maximize revenue opportunities. Implementing these recommendations can enhance operational efficiency, support data-driven decision-making, and contribute to long-term business growth.

Overall, this project demonstrates how data analytics and business intelligence tools can be utilized to improve supply chain visibility, identify operational challenges, and drive strategic business outcomes.